

Manoj Kumar Thapa

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UK Graduate Route visa valid to 2028 – no sponsorship required

EDUCATION

MSc Artificial Intelligence, Aston University, UK – 70.37% Jan 2025 – Jul 2026

BE Computer Science, Dr. Ambedkar Institute of Technology, India – CGPA 9.45/10 Aug 2017 – Sep 2021

PROJECTS

Planck AI – Agentic Research Assistant | *Python, FastAPI, React, Docker* | [GitHub](#) · [Demo](#) Sep 2025 – Nov 2025

- Built an LLM agent in Python without an agent framework, orchestrating 4 tools (web search, code execution in 7 languages, PDF/URL reading, image analysis) across up to 8 planning steps.
- Designed mid-run failover across Groq, Gemini and NVIDIA behind one OpenAI-compatible interface, sustaining **30/30** task completion under fault injection with the primary provider disabled on every call.
- Built an evaluation harness with a 30-question golden set and a rubric committed to git before the first run, measuring **80%** correct tool selection, **2.7 s** median latency and **zero** incorrect answers.

CogniGraph – Graph RAG Explorer | *Python, ChromaDB, NetworkX, OCR* | [GitHub](#) · [Demo](#) Jun 2025 – Jul 2025

- Built a knowledge-graph RAG app that turns documents, including scanned PDFs via OCR, into an interactive 3D graph and answers only from passages retrieved from a ChromaDB vector database.
- Benchmarked retrieval at **95% hit@3** against a keyword baseline; found self-written test questions overstated accuracy by **35 points** and rebuilt the set in natural user phrasing to remove the bias.
- Diagnosed silent truncation in the embedding model past **~1,400 characters**, which dropped text from larger chunks with no error raised, and set chunk size from that measurement.

FruitGuard AI – MSc Dissertation | *PyTorch, YOLOv8, ONNX Runtime* | [Demo](#) Aug 2025 – Apr 2026

- Fine-tuned a YOLOv8 computer-vision model with a two-phase transfer-learning pipeline to **96.3%** classification accuracy from only **207** labelled images, a **95%** reduction in annotation data.
- Optimized inference with ONNX Runtime to real-time **73 FPS**, deployed as a live Hugging Face demo.

NeuroArc – AI Job Application Assistant | *FastAPI, React, Groq API, OCR* | [GitHub](#) · [Demo](#) Nov 2025 – Dec 2025

- Built an end-to-end assistant that parses CVs (PDF/DOCX, OCR fallback), pulls live UK jobs from the Reed API, scores CV-job fit and exports a tailored CV as PDF.
- Prevented hallucinated matches: the LLM returns requirement-evidence pairs as structured JSON, and code computes the weighted score only from evidence verified against the CV text.

EXPERIENCE

Mentor, MSc Artificial Intelligence, Aston University | *Birmingham, UK* Aug 2025 – Jan 2026

- Mentored **10+** MSc AI students in Python and machine learning through weekly one-to-one sessions.

Software Engineer, Accenture | *Bangalore, India* Oct 2021 – May 2022

- Cut REST API latency **40%** by eliminating N+1 queries and optimizing SQL in Java Spring Boot services.
- Built and maintained production REST API services for a large-scale enterprise application, implementing business logic and PostgreSQL/MySQL data-access layers.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, TypeScript

LLMs & Agents: Tool/function calling, agent loops, LangGraph, RAG, prompt engineering, structured JSON output, multi-provider LLM fallback (Groq, Gemini, NVIDIA; OpenAI-compatible APIs)

Retrieval & Data: ChromaDB (vector database), embeddings, semantic search, OCR, PostgreSQL, MySQL

ML & Computer Vision: PyTorch, YOLOv8, transfer learning, fine-tuning, ONNX Runtime, quantization

Evaluation: Golden-set design, fault injection, baseline controls, latency benchmarking

Backend & Frontend: FastAPI, Pydantic, Spring Boot, REST API design, React, Tailwind CSS

Tools & Deployment: Docker, Docker Compose, Git, Hugging Face Spaces, Render

Certifications: [AI Agents Intensive \(Google, Kaggle\)](#) | [Machine Learning Specialization \(Stanford\)](#) |

[Deep Learning Specialization \(DeepLearning.AI\)](#)